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Connecting Environments to Human Health: A Research-Based Curriculum Module

Abstract

Environmental health research explores how climate change, pollution, and environmental injustice affect people's health, and then develops solutions. This lesson uses Harvard T.H. Chan School of Public Health's Environment and Climate Health research area as a map of the field, with two recent examples: predicting the health impacts of new data centers, and protecting people from wildfire smoke.

Introduction

The air we breathe, the water we drink, and the places we live and work all shape our health. Environmental health is the science of those connections. At Harvard Chan School, researchers in this field study how climate change, pollution, and environmental injustice affect health, and they design solutions. Their work spans many labs and departments, from environmental health and global health to biostatistics. Recent topics range from extreme heat and workers' health, to PFAS chemicals in veterans' blood, to wildfire smoke and the energy demands of data centers. This lesson explores how researchers connect environments to health, and how their findings can guide decisions that protect communities.

How Do Scientists Study This?

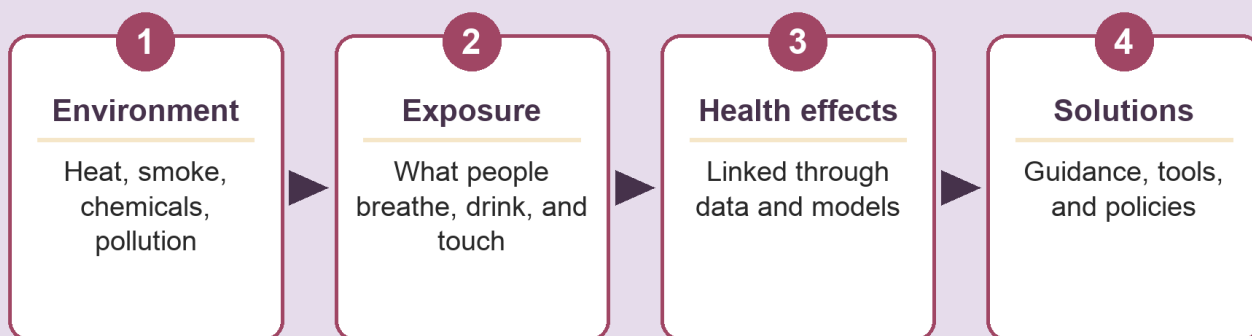
Environmental health researchers use many methods. Exposure scientists measure what people come into contact with, such as pollutants in air or chemicals in blood. Epidemiologists and biostatisticians use large datasets to link exposures to health outcomes. Engineers and data scientists build models to predict future impacts. For example, a new project co-led by biostatistician Francesca Dominici and electrical engineer Le Xie is building a three-part framework for data centers: a data pipeline tracking the energy use and emissions of existing centers, models estimating how pollution from them might harm nearby communities, and an interactive toolkit for planning where new data centers should go.



FROM THE ARTICLE

Aerial view of a large data center campus in Hutto, Texas. Photo: BackyardProduction / iStock.

How environmental health research works



Study the Process

Measure exposure

Scientists track what people encounter, from air pollution and heat to chemicals in blood.

Link to health

Statistical models connect exposures to illnesses and other health outcomes.

Guide decisions

Tools and guidance help communities, regulators, and developers protect health.

What Did They Discover?

Environmental health research is turning findings into practical guidance. The data center project aims to give communities, regulators, utilities, and developers usable information about a proposed center's impacts, as resistance grows over strains on power grids and pollution. It was one of 20 projects funded through Harvard's Frontiers of Innovation for Societal Impact Fund. Another example is wildfire smoke: in summer 2026, smoke degraded air quality across parts of Canada, the U.S., and other regions. Harvard Chan and C-CHANGE experts explained how smoke affects the lungs, heart, and body, who is most at risk, and how to reduce exposure through better indoor air filtration, portable air cleaners, and protective masks.

Data centers and health

3

parts in the framework: data, health models, toolkit

20

projects funded by Harvard's societal impact fund

3

smoke protections: filtration, air cleaners, masks

Dominici described the data center project as an effort to tackle a contentious issue from a scientific, rigorous, and unbiased perspective.

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Takeaways

Environment shapes health

Climate change, pollution, and injustice all affect people's health.

Many sciences, one goal

Exposure science, statistics, and engineering work together to protect health.

Research guides action

Toolkits and guidance help communities make healthier decisions.

Key Terms

Environmental health

The field of science that studies how the environment, including air, water, and climate, affects human health.

Exposure

Contact with something in the environment, such as a pollutant or extreme heat, that may affect health.

Environmental injustice

When some communities, often low-income or communities of color, bear an unfair share of environmental harm.

PFAS

A group of long-lasting human-made chemicals, sometimes called forever chemicals, found in many products and in people's blood.

Key Terms continued

Air filtration

Removing particles and pollutants from indoor air, for example with filters or portable air cleaners.

Activity

Environmental Health Detectives

Students investigate one environmental exposure in their community and plan how researchers could study it.

Materials: Exposure cards: wildfire smoke, extreme heat, traffic pollution, a proposed data center; Research plan worksheet; Internet access (optional)

1. Choose an exposure card and describe how it could affect the lungs, heart, or other parts of the body.
2. Decide how you would measure people's exposure, such as sensors, surveys, or blood tests.
3. Identify which groups might be most at risk and why.
4. Propose one protective action for individuals and one for a school or workplace.
5. Present your plan and explain which kinds of scientists would need to work together on it.

References

Harvard T.H. Chan School of Public Health. Environment & Climate Health.
<https://hsph.harvard.edu/topic/environment-climate-health/>

Feldscher, K. (2026). Predicting data centers' energy and health impacts. Harvard Chan School.
<https://hsph.harvard.edu/news/predicting-data-centers-energy-and-health-impacts/>

Roache, C. (2026). Wildfire smoke: The health and environmental impacts. Harvard Chan School.
<https://hsph.harvard.edu/news/wildfire-smoke-the-health-and-environmental-impacts/>